

A modular causal adaptation architecture for machine learning on data-scarce rare disease

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Abstract

Rare diseases are called orphan diseases because each affects few patients, leaving them data-scarce to model. Context-conditioned perturbation models cannot parameterize a disease context absent from pretraining without additional adaptation, and a patient-derived line carrying a particular lesion will never be screened at the scale such models require. Rather than learning rare-disease biology from scratch, this work isolates the pathological shift within a pre-wired model of general biology, parameterizing the mutation as a rank-one deviation added to a frozen structural causal model. The factorization reduces estimation from quadratic to linear in the number of genes, and a soft one-hot vector names the network entry node. A modular loop adds counterfactual scoring, bootstrap ensemble uncertainty and active experiment selection. On a simulated 50-node network, localization rises from 0.00 to 0.78 as observations increase from 5 to 40 (chance 0.02), while an unconstrained sparse deviation under identical regularization never exceeds 0.11. On real protein-signalling data recovery remains near chance. Controlled experiments with known injected lesions place that failure in identifiability rather than in the estimator; we report this identifiability boundary as the primary contribution.

1 Introduction

Roughly seven thousand rare diseases are recognised, and the economics of each are the same: few patients mean few trials, few trials mean little data, and little data means weak incentive to invest. Individually these conditions are negligible; collectively they are not. Models trained across large perturbation screens have learned a great deal about how cells respond to intervention, but the way that knowledge is stored blocks its transfer. Context-conditioned perturbation models represent the cell line, tissue or genetic background through learned identifiers, so prediction is restricted to combinations seen in training [1]. The context that matters for a rare disease, a patient-derived line carrying a specific lesion, has never been screened at that scale and never will be, and training a disease-specific model from scratch is not an alternative because the data do not exist.

We therefore ask a narrower, directly verifiable question: given a handful of observations from a context never seen in pretraining, can a model identify where in the network the disease enters? We call this lesion localization. It is weaker than predicting a treatment but checkable against ground truth. The lesion studied is a structured single-lesion rewiring, one node acquiring a new set of downstream targets. This describes a fusion oncogene well and protein misfolding or a pure loss-of-function allele poorly; we do not claim a universal rare-disease adapter.

2 Contributions.

1. A rank-one causal adapter that models a disease as a low-rank deviation from a frozen structural causal model, reducing estimation from $O(G^2)$ to $O(G)$ parameters and returning an interpretable output, the network entry node.
2. Evidence that the inductive bias, not the sparsity penalty, makes few-shot recovery possible, and an identifiability boundary that attributes failure on a small real benchmark to identifiability rather than to the estimator.
3. A negative result reported in full: under a misspecified causal order reliability collapses while structural metrics give no warning.

3 Method

3.1 Frozen causal foundation

The prior is a non-linear structural causal model over G genes. Under a fixed topological ordering W_0 is strictly triangular and

$$x_i = \tanh(a_{c,i} \sum_{j < i} W_{0,ji} x_j + b_i) + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2), \quad (1)$$

where the gain vector $a_c \in \mathbb{R}^G$ modulates each gene's sensitivity, so contexts share wiring but differ in responsiveness; $\tanh$ is a generic saturating nonlinearity, not a kinetic model. An intervention $do(x_i = v)$

replaces gene i 's equation and leaves all others intact, which makes propagation causal rather than predictive. Because W_0 is triangular, Eq. 1 is solved exactly by $d + 1$ fixed-point iterations, d the longest path.

3.2 Rank-one causal adapter

The foundation is frozen. For a novel context we learn only the deviation

$$W_{\text{lesion}} = W_0 \odot a_c + u v^T, \quad u = \text{softmax}(\ell / \tau), \quad L = \text{MSE} + \lambda \|v\|_1. \quad (2)$$

u is a soft one-hot over the node the lesion enters through and v the sparse rewiring it causes; τ is annealed from 2.0 to 0.2 so that optimization first explores candidate entry nodes and then commits. A routed bias $\delta b = u \odot s$ lets the same form express a change in a node's level as well as in its downstream influence. The lesion has an entry point and a structured downstream effect, hence $2G$ parameters instead of G^2 , which makes estimation from $k \approx 10$ observations well posed, and $\arg \max u$ is a single named gene rather than a dense matrix requiring post-hoc attribution.

3.3 Counterfactual scoring, uncertainty and acquisition

Every candidate intervention is propagated through the adapted graph and scored against a phenotype readout. Six adapters are fitted to bootstrap resamples; their spread is the epistemic signal. Acquisition selects the perturbation on which members disagree most; it is run, appended and the adapter refit, closing the loop (Fig. 1).

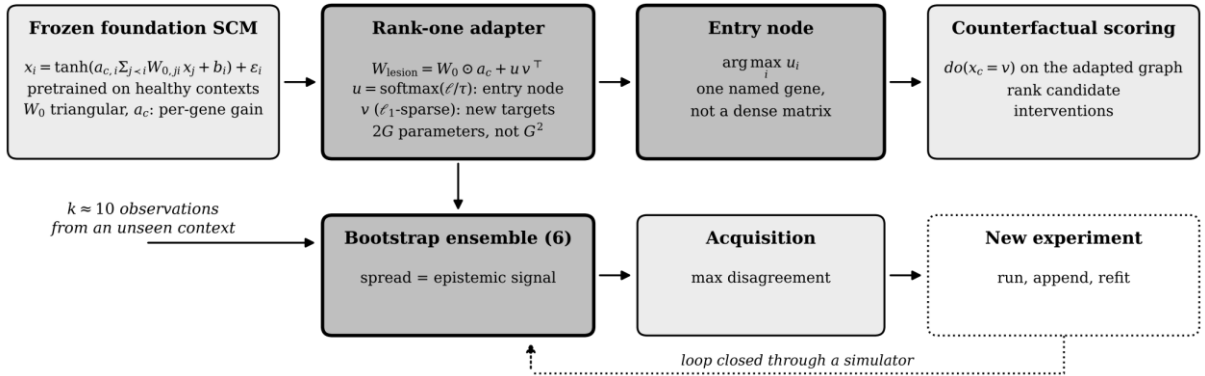


Figure 1. The modular loop. Light: pretrained and frozen. Dark, heavy outline: fitted on k observations from the novel context. Dotted: closed through a simulator, not a laboratory.

4 Experimental setup

Simulated networks. Random DAGs over G genes, edge probability 0.06, weights $N(0, 0.9)$. Ten contexts differ by gain vectors; the pretraining corpus contains unperturbed observations plus single- and double-gene knockdowns per context. A disease gives an anchor gene six new outgoing edges.

Phenotype. The readout loads on deep descendants of the lesion, excluding its direct targets, and candidate interventions are restricted to genes outside it, to prevent leakage.

Real data. Sachs et al. [3]: 11 phospho-proteins, 7,466 single cells, nine reagent conditions. Each reagent acts on a known protein, so hiding its identity gives real ground truth. We hold out one regime at a time, fit the foundation on the rest and reveal k observations of the held-out regime (chance 1/11).

Baselines. *No adapt*: the frozen foundation. *Dense*: an unconstrained $G \times G$ deviation under identical ℓ_1 . *Max shift* (real data): the protein deviating most from the unperturbed mean.

5 Results

Localization. The foundation recovers the true graph with precision 0.93 and recall 0.91. Accuracy rises from 0.00 at $k = 5$ to 0.78 at $k = 40$ (Fig. 2a). The dense comparison is the informative one: with the same regularization and data it never exceeds 0.11, so sparsity alone does not make the lesion recoverable; the factorized structure does. Edge F1 of the rewiring follows the same pattern (0.53 vs 0.28).

Active selection. From eight observations, maximum-disagreement acquisition reaches perfect localization in one round and holds it for four, while random selection oscillates (Fig. 2b; one lesion, one seed). Ensemble spread falls from 0.017 to 0.005 as accuracy rises.

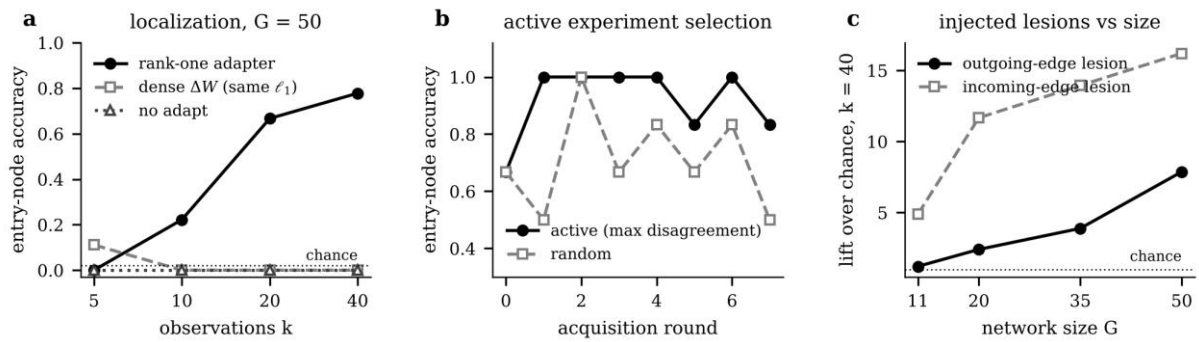


Figure 2. Simulated networks, as in the workshop abstract. (a) Entry-node accuracy for one injected fusion-like lesion, $G = 50$, three seeds. (b) Active vs random acquisition from eight observations. (c) Recovery of known injected lesions as a multiple of chance, $k = 40$.

Real data. On Sachs, leave-one-regime-out foundations recover 0.72–1.00 of consensus edges (precision 0.39–0.50). Localization nonetheless fails: at $k = 50$ rank-one reaches 0.10 against chance 0.09, dense 0.00, and max shift, which uses no model, 0.29 (Fig. 3a). Signal is confined to two regimes, psitectorigenin acting on PIP2 and PMA acting on PKC (Fig. 3b). On this benchmark the method is beaten by a one-line baseline; it validates the mechanism, not the rare-disease claim.

Diagnosis: an identifiability boundary. Injecting known lesions into networks of increasing size separates a wrong adapter from an unidentifiable problem (Fig. 2c). Lift over chance grows from 1.2 at $G = 11$ to 7.9 at $G = 50$, and incoming-edge lesions are two to five times more recoverable than outgoing ones, since incoming edges set a directly observed value.

Misspecified causal order. Reversing 1.8% and 7.1% of the supplied order lowers mean accuracy from 0.56 to 0.37 and 0.30, still about fifteen times chance, but the standard deviation grows from 0.00 to 0.21 while edge precision falls only from 0.91 to 0.78 (Fig. 3c). A foundation can look healthy by every structural metric while sending the adapter to the wrong node.

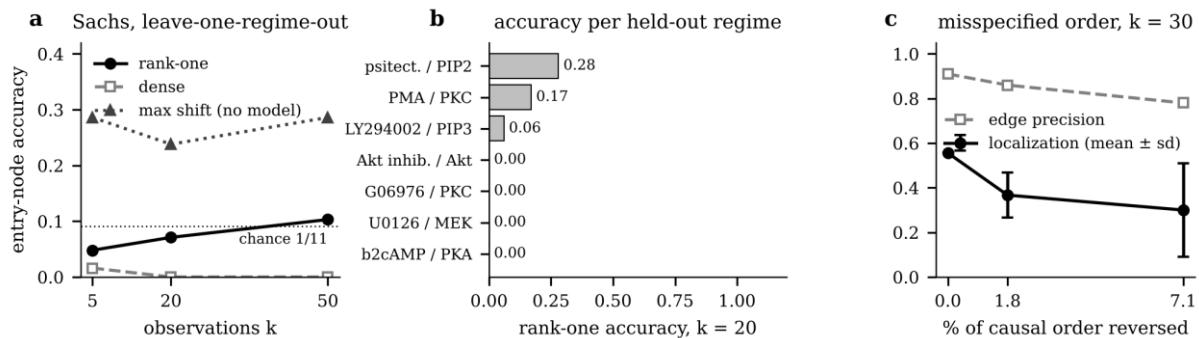


Figure 3. Real data and robustness. (a) Sachs, leave-one-regime-out localization. (b) Rank-one accuracy per held-out regime, $k = 20$. (c) Corrupting the supplied causal order, $G = 50$, five draws per level.

6 Discussion and limitations

A rank-one deviation from a frozen causal prior is recoverable from tens of observations, and the factorization rather than sparsity is what makes it so. The identifiability boundary is a more useful output than any single accuracy, because it states a design requirement: the disease context must be perturbed enough, and the lesion must be visible in the readout. Genome-scale screens [2] are the natural setting. The ordering is supplied rather than discovered, and the method cannot detect when it is wrong. All models are acyclic while signalling contains feedback; tanh is generic; the loop closes through a simulator; and the rank-one form encodes one lesion topology. No claim is made about therapeutic relevance or patient tissue.

7 Future directions

Generalizing the lesion prior: rank- k deviations UV^T for interacting lesions, an attenuation mode for loss of function, and variant-level parameterization.

Characterizing identifiability: map the probability of correct localization over coverage, network size, lesion topology and noise, so a practitioner knows in advance whether the question is answerable.

Closing the diagnostic gap: find a statistic that warns when the causal order is wrong (held-out interventional likelihood, entropy over u), and acquire directly on entropy over u .

Validation: an inducible EWS::FLI1 knockdown in Ewing sarcoma cells (the fusion is a known rank-one anchor) and multi-donor iPSC CRISPRi screens (each donor an unseen context).

8 Conclusion

We framed rare-disease modelling as estimating a low-rank deviation from a frozen causal model of general biology rather than learning disease biology from scarce data. The rank-one adapter localizes an injected lesion well above chance, the factorization rather than the regularization carries the result, failure on a small real benchmark is located in identifiability, and reliability under a misspecified order collapses invisibly to structural metrics. We regard the boundary as the contribution: it states what a deployment needs, in units a subsequent study can test.

Github - <https://github.com/nandinii27/A-modular-causal-adaptation-architecture-for-machine-learning-on-data-scarce-rare-disease>

References

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